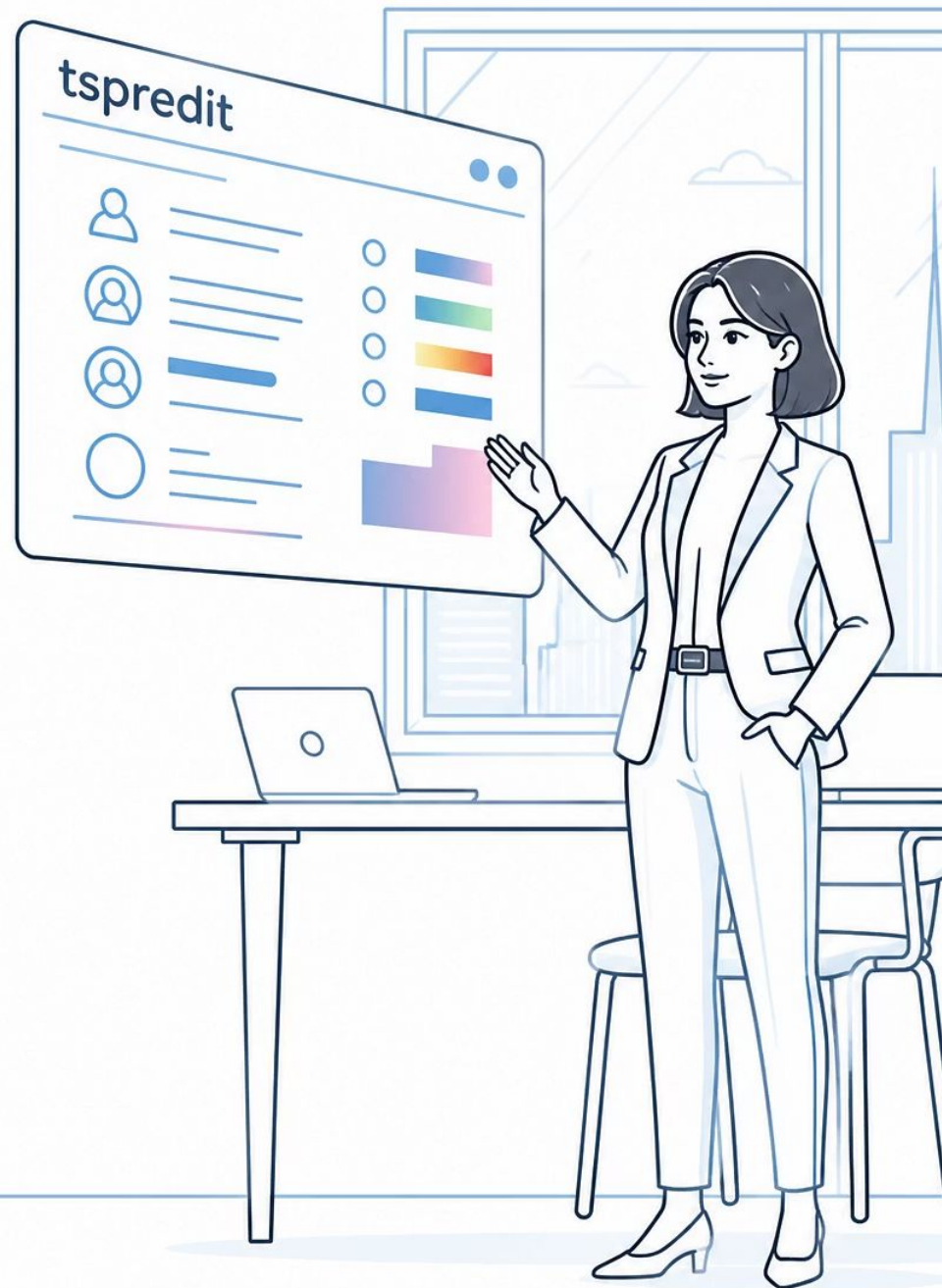




Time Series Forecasting in R: Integrated Preprocessing and TSPred IT

The Challenge in Time Series Forecasting

Time series forecasting is a fundamental challenge — from demand forecasting to financial modeling. Despite advances in predictive models, a critical gap persists in existing frameworks.

Current tools focus almost exclusively on hyperparameter optimization. Preprocessing is treated as a separate manual step. This fragmented approach ignores the interaction between preprocessing choices and model performance.

Critical Need

Simultaneous optimization of preprocessing pipelines and model architectures

Current Gap

Frameworks optimize models but ignore the tuning of preprocessing parameters

Main Question

Which combination of preprocessing and model maximizes forecasting accuracy?

Integrated tspredit Pipeline

01

Raw Data

Input time series

02

Preprocessing

Filtering and normalization

03

Augmentation

Data expansion

04

Training

Model fitting

05

Evaluation

Metrics and optimization

Each step is co-optimized by `ts_integ_tune`.

Preprocessing Filters

tspredict offers **18 filters** for smoothing, noise removal, and seasonal adjustment. Uniform interface: `fit/transform`.

Time Domain

MA, EMA, SES, QES, HP, LOWESS,
Spline, Recursive, Winsorization,
Smooth

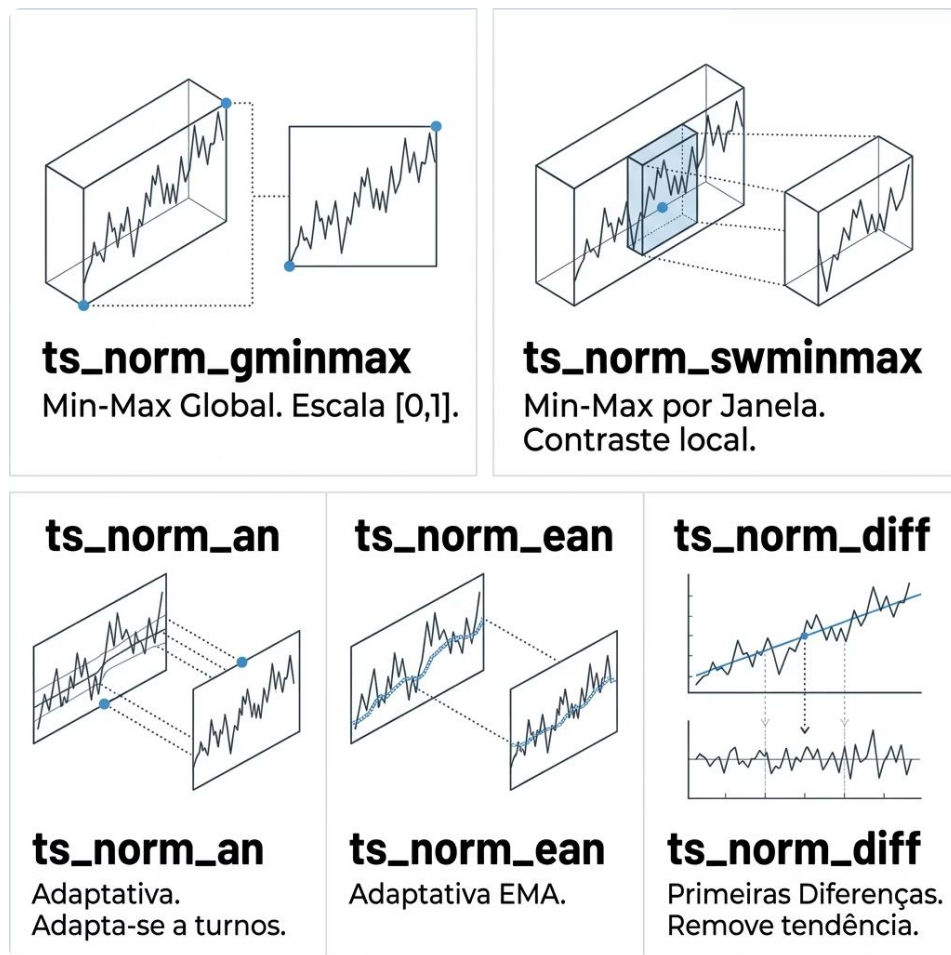
Frequency Domain

FFT, Wavelet (haar, d4, la8, bl14, c6)

Empirical Decomposition

EMD, Robust EMD (CEEMD), Seasonal
Adjustment (BATS), Kalman

Data Normalization



Normalization Strategies

Six approaches — from global scaling to dynamic adaptation — for stationary and non-stationary time series.

- ❏ Adaptive normalization (`ts_norm_an`) was proposed by Ogasawara et al. (IJCNN 2010). It is especially effective for series with distribution shifts over time.

All normalizations support outlier removal via `outliers_boxplot()` and implement `inverse_transform` to recover the original scale in predictions.

Prediction Models

Statistical Models

ARIMA via `auto.arima` — automatic order selection by information criterion.

Kernel & Ensemble

SVM (`e1071`) with linear, radial, and polynomial kernels.
Random Forest (`randomForest`) for variance reduction.

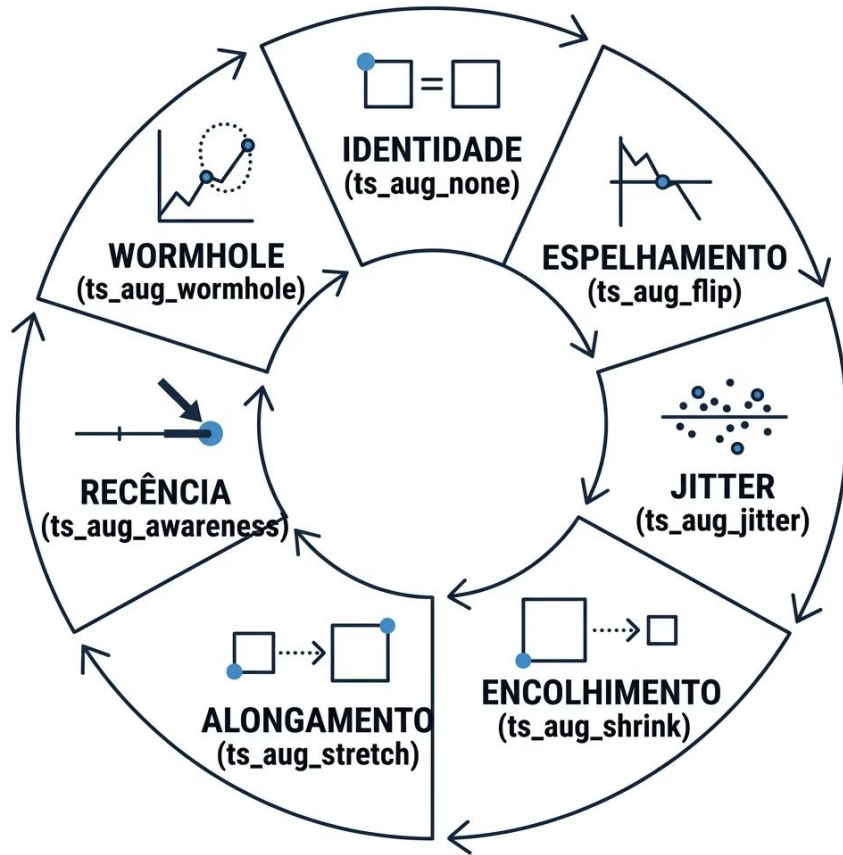
Neural Networks

MLP (`nnet`) and **ELM** (`elmNNRcpp`) — non-linear mappings with sliding windows.

KNN

K-Nearest Neighbors — regression by similarity in the lag space. Normalization recommended for distance metrics.

Data Augmentation



Training Set Diversification

The `tspredict` implements **7 augmentation strategies** over sliding windows. The goal: expand sample diversity and reduce overfitting.

→ Scale Preservation

Flip, Shrink, and Stretch maintain the window mean, altering only the variance.

→ Recency

Awareness and AwarenessSmooth prioritize recent observations with progressive smoothing.

→ Temporal Permutation

Wormhole replaces lags with older values, generating plausible alternative trajectories.

Integrated Tuning: ts_tune and ts_integ_tune

ts_tune
Adjusts input_size and
model hyperparameters.

ts_integ_tune
Integrated search over
preprocessing and
augmentation.



Cross-Validation
Evaluates performance in
a robust and reliable way.

ts_tune

- Search over input_size and hyperparameters
- Cross-validation with configurable folds
- Selects the configuration with the lowest mean error

ts_integ_tune

- Extends ts_tune with search over preprocessors
- Includes augmentation in the training loop
- Full co-optimization of the pipeline

Evaluation Metrics

MSE

Mean Squared Error

`MSE.ts(actual, prediction)` — penalizes large errors.

sMAPE

Symmetric Mean Absolute Percentage Error

`sMAPE.ts(actual, prediction)` — scale-independent. Used in M3/M4.

R²

Coefficient of Determination

`R2.ts(actual, prediction)` — measures explained variance.

📄 All metrics follow the `evaluate(model, output, prediction)` interface from **daltoolbox**. Ensures consistency across experiments.

Benchmarks and Real Data

Forecasting Competitions

M1 / M3 / M4

Makridakis (1982–2018).
Monthly, quarterly,
yearly, and daily series.

NN3 / NN5

111 monthly business
series (NN3). 111 daily
ATM series (NN5).

CATS / Santa Fe / EUNITE

Artificial series, chaotic laser, and electric load (30-
min resolution).

Real Data (FAOSTAT & Ipea)

Environment

Bioenergy, Climate
(NASA-GISS), Emissions
(GHG), Fertilizers,
Pesticides.

Economy

GDP (FAOSTAT). Daily
Ipeadata: 12 series, up
to 8,154 obs. Monthly:
23 series.

Finance

50 stocks from B3 (IBOVESPA). Daily OHLCV data —
2025.

Connection with daltoolbox

Class Inheritance

`ts_reg` and `ts_regsw` extend the base classes of `daltoolbox`. This enables `fit()`, `predict()` and `evaluate()` in a uniform way.

Experimental Interface

Reproducible experiments via `ts_sample()`, `ts_projection()` and `loadfulldata()`. Fully integrated into the DAL ecosystem.

Experiment Pipelines

Rolling-origin evaluation with multiple horizons (`steps_ahead`) and offsets. Compatible with the CEFET/RJ DAL protocols.

Complete Usage Example

Data Preparation

```
library(daltoolbox)
data(tsd)

# 1. Sliding windows
ts <- ts_data(tsd$y, 10)

# 2. Train/test split
samp <- ts_sample(ts, test_size = 5)
io_train <- ts_projection(samp$train)
io_test <- ts_projection(samp$test)
```

Fitting and Prediction

```
# 3. Integrated fitting
tune <- ts_integ_tune(
  input_size = 3:5,
  base_model = ts_elm(),
  ranges = list(nhid = 1:5),
  preprocess = list(ts_norm_gminmax()),
  augment = list(ts_aug_jitter())
)
model <- fit(tune,
  x = io_train$input,
  y = io_train$output)

# 4. Prediction and evaluation
pred <- predict(model,
  x = io_test$input[1,],
  steps_ahead = 5)
evaluate(model,
  as.vector(io_test$output),
  as.vector(pred))
```

What happens internally

01

ts_data

Sliding windows with $t\{\text{lag}\}$ columns

02

ts_sample + ts_projection

Splits train/test. Projects into (X, y)

03

ts_integ_tune → fit

Integrated CV search

04

predict + evaluate

MSE/sMAPE/ R^2 on original scale

Why TSPred-IT Stands Out

Full Pipeline Automation

Eliminates manual interventions between preprocessing, training, tuning, and evaluation.

End-to-end reproducibility ensures consistent results across all experiments.

Innovative Co-Optimization

First framework to unify preprocessing parameters and hyperparameters in a single search space.

Captures critical interactions between data transformations and algorithms.

Extensible Architecture

Plugin system supports custom methods, models, and metrics.

Available in R and Python with open APIs for community contributions.

Comprehensive Model Support

Unified interface for statistical, machine learning, and deep learning models.

Combine algorithms without changing the workflow.

Proven Real-World Performance

Validated on economic, environmental, and industrial data.

Competitive or superior accuracy compared to specialized tools.

Conclusion and Future Directions

TSPred-IT fills the critical gap between preprocessing and model optimization in time series forecasting

Key Takeaways

- Automates the complete forecasting pipeline
- First tool with joint preprocessing and model optimization
- Supports statistical, ML, and deep learning approaches
- Proven accuracy improvements on real-world data
- Available on CRAN and GitHub

Future Improvements

- More deep learning integration (Transformers, attention)
- Advanced search: Bayesian optimization and evolutionary algorithms
- Multivariate time series support
- Distributed computing for large-scale experiments
- AutoML for non-expert users

Acknowledgements



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